



IoT Smart Home Monitoring

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1. Project Overview

2.1 Project Idea

The project presents a Smart Home Sensor Monitoring System that continuously monitors environmental conditions inside a home. The system focuses on collecting temperature and light intensity data using sensors connected to an ESP32 microcontroller.

The collected data is transmitted wirelessly to a cloud platform, where it is stored and visualized in real time. This allows users to monitor indoor conditions remotely and supports smart home automation concepts.

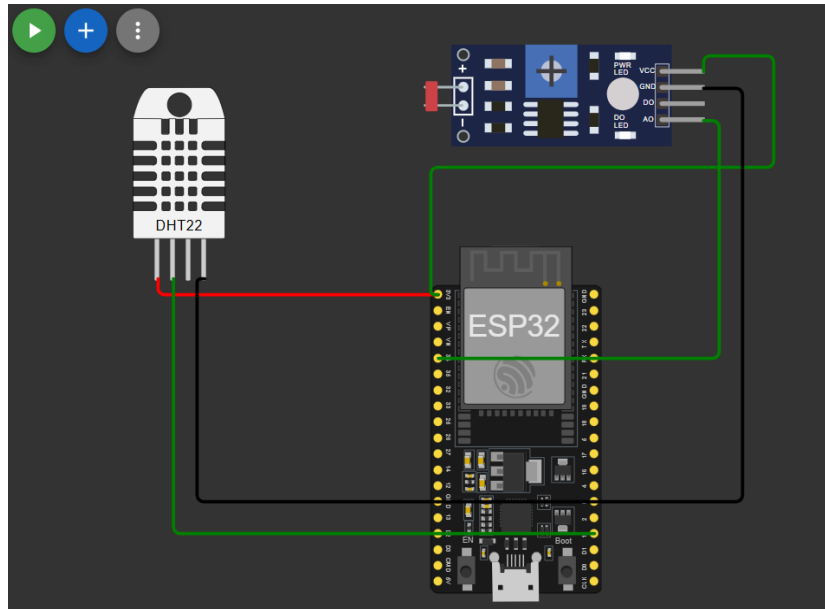
2.2 Project Objective

The main objectives of this project are:

- To monitor ambient temperature inside a smart home environment using a digital temperature sensor.
- To measure light intensity levels using a photoresistor (LDR) sensor.
- To transmit the collected sensor data to a cloud-based IoT platform using WiFi communication.
- To visualize the sensor data using graphs and charts that help in understanding environmental trends and supporting decision-making in smart home systems.

2 . Deliverable 1 – IoT Network Design and Visualization

2.1 System Architecture



The IoT system is designed using an **ESP32 microcontroller** as the main processing and communication unit. The ESP32 is connected to multiple sensors that collect environmental data from the surrounding environment.

The system uses **WiFi communication** to send the collected data to the **ThingSpeak cloud platform**, where the data is stored and visualized in real time.

The overall architecture consists of:

- ESP32 microcontroller
- Temperature sensor (DHT22)
- Light sensor (LDR)
- WiFi network
- Cloud platform (ThingSpeak)

2.2 Sensors Description

2.2.1 Temperature Sensor (DHT22)

The DHT22 is a digital temperature sensor used to measure the **ambient temperature**. It provides accurate and stable readings, which makes it suitable for smart home applications.

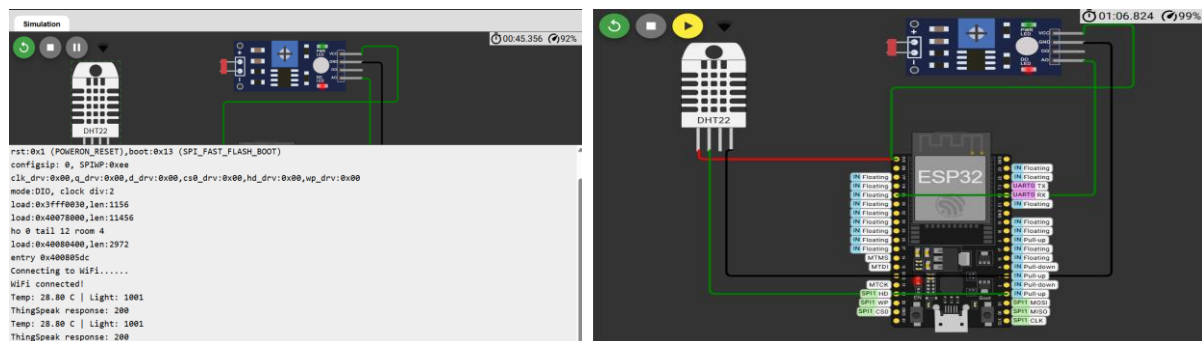
In this project, the DHT22 sensor is connected to the ESP32 through a digital input pin. The sensor continuously measures the temperature and sends the readings to the microcontroller for further processing.

2.2.2 Light Sensor (LDR)

A Light Dependent Resistor (LDR) is used to measure **light intensity** in the environment. The resistance of the LDR changes according to the amount of light falling on it.

The sensor is connected to an analog input pin of the ESP32, allowing the system to read varying light levels. This sensor is suitable for smart home systems where automatic lighting control may be required.

2.3 Data Collection Process



The ESP32 reads data from the connected sensors at regular time intervals. The temperature value is obtained from the DHT22 sensor, while the light intensity value is read as an analog signal from the LDR sensor.

The system is programmed to collect and transmit data every **15 seconds**.

If a sensor reading fails, the system prints an error message on the serial monitor to indicate a connection or reading issue.

2.4 IoT Communication and Cloud Platform

The collected sensor data is transmitted to the **ThingSpeak IoT platform** using HTTP requests over WiFi.

Each sensor value is assigned to a specific field in the ThingSpeak channel:

- **Field 1:** Temperature (°C)
- **Field 2:** Light Intensity

ThingSpeak stores the incoming data and provides tools for real-time data

3.5 Data Visualizations

2.5.1 Temperature Visualization

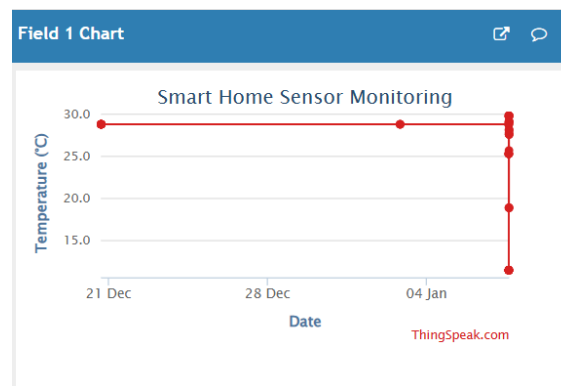


Figure 1: Temperature Chart

A line chart is used to display temperature readings over time.

This visualization helps in monitoring temperature changes and detecting abnormal increases or decreases inside the smart home environment.

2.5.2 Light Intensity Visualization

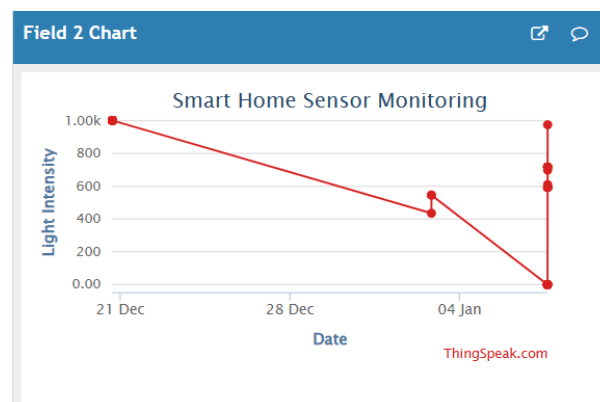


Figure 2: Light Chart

Another line chart is used to represent light intensity values over time. This visualization allows users to understand lighting patterns and identify periods of high or low light levels.

2.5.3 Scatter Plot (Temperature vs Light Intensity)

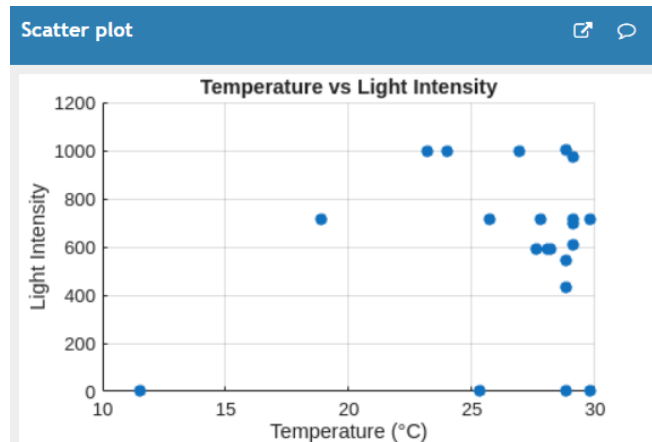


Figure 3: Scatter_Temp_vs_Light

A scatter plot is used to analyze the relationship between temperature and light intensity. The visualization shows that higher light intensity is often associated with higher temperature values, indicating a correlation between the two environmental factors.

2.5.4 Dual-Axis Plot

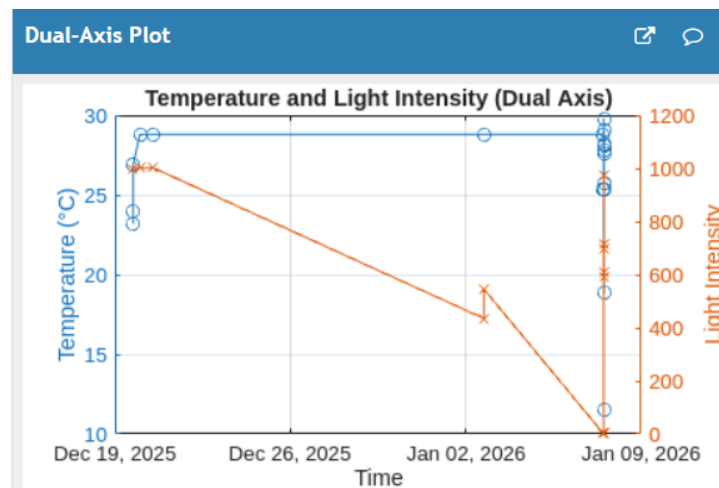


Figure 4 : DualAxis_Temp_Light

A dual-axis plot is used to display both temperature and light intensity on the same graph over time.

This visualization enables easy comparison between the two variables and provides a clearer understanding of their behavior and relationship.

2.6 Wokwi Project Link

The complete IoT simulation for this project is available at the following link:

🔗 <https://wokwi.com/projects/450140818799747073>

3. Deliverable 2 – Data Collection, Visualization, and Analysis

3.1 Dataset Description

- **Dataset:** Sensor readings collected from the project: **Temperature** from DHT22 and **Light Intensity** from LDR.
- **Source / Obtained from:** Data is collected directly from the **ESP32** during project operation or from the **Wokwi simulation**.
- **Representation:** Each record includes:
 - Temperature (°C)
 - Light Intensity
 - Timestamp of the reading

3.2 Chosen Visualizations and Justification

- **Line Charts (Field 1 & Field 2):**
 - Display temporal changes of each sensor.
 - **Relevance:** Allows easy monitoring of environmental changes and detection of abnormal readings.
- **Scatter Plot (Temperature vs Light Intensity):**
 - Examines the relationship between the two variables.
 - **Relevance:** Shows whether higher light intensity is correlated with higher temperatures.
- **Dual-Axis Plot:**
 - Displays both variables on the same graph.
 - **Relevance:** Provides a clear comparison and understanding of trends.

3.3 Chosen Action and Proof (Screenshot)

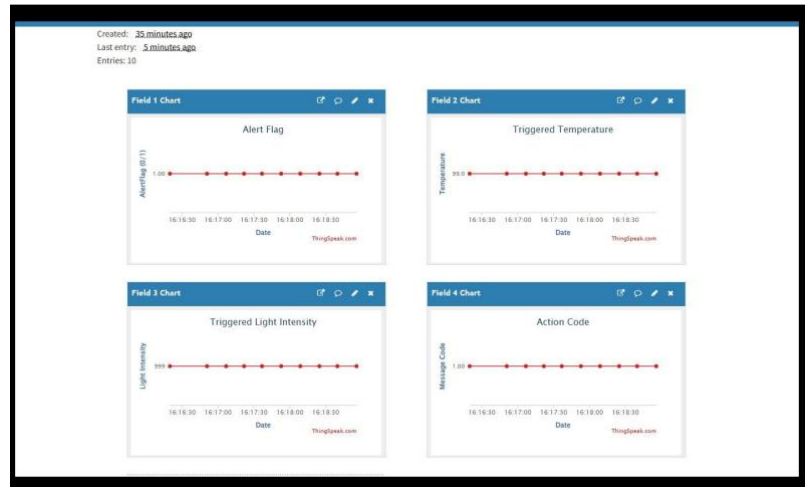


Figure 5: ChosenAction_ThingSpeakProof

- **Action:** Sending sensor data to **ThingSpeak** and visualizing it using the platform.
- **Proof:**
 - Serial Monitor screenshot (2.3 Data Collection Process)
 - Field 1 Chart – Temperature
 - Field 2 Chart – Light Intensity
 - Scatter Plot
 - Dual-Axis Plot

3. 4 How the Channel Supports the Project Idea

- The ThingSpeak channel receives data **in real-time**, allowing users to:
 - Monitor **temperature and light intensity** in the smart home.
 - Detect sudden changes in the environment.
 - Analyze trends through visualizations to support **smart home monitoring**.

Deliverable 3 – Analysis, Discussion, and Future Work

1. Data Analysis and Discussion

- Collected sensor data (Temperature and Light Intensity) was analyzed using **ThingSpeak visualizations**.
- **Observations:**
 - Temperature tends to increase when light intensity is high, indicating a correlation between the two variables.
 - Both sensors provide stable readings, confirming the system works reliably.
- Analysis is supported by:
 - Line charts (Field 1 & Field 2)
 - Scatter plot
 - Dual-axis plot

2. Key Findings

- The system successfully monitors environmental conditions in real-time.
- IoT cloud integration allows remote monitoring and easy access to historical data.
- Visualizations help in detecting abnormal changes in the environment.

3. Future Work / Recommendations

- Add more sensors (e.g., humidity, motion, air quality) for a complete smart home solution.
- Implement automatic alerts when readings exceed thresholds.
- Integrate with mobile apps for user-friendly notifications.

4. References

- <https://www.adafruit.com/product/385>
- <https://github.com/mathworks/thingspeak-arduino>
- https://www.espressif.com/sites/default/files/documentation/esp32_technical_reference_manual_en.pdf